

Simon's Algorithm

Simon's Problem

Input $\rightarrow f: \Sigma^r \rightarrow \Sigma^m$

Promise $\rightarrow$ There exists $s \in \Sigma^r$ such that

$$[f(x) = f(y)] \iff [(x=y) \text{ or } (x \oplus s = y)]$$

Ex. 1

x	$f(x)$
000	10011
001	00101
010	00101
011	10011
100	11010
101	00001
110	00001
111	11010

$$S = 011$$

$$f(000) = f(011) = 10011$$

$$f(001) = f(010) = 00101$$

$$f(100) = f(111) = 11010$$

$$f(101) = f(110) = 00001$$

$$P(y) = \frac{1}{2^n} \rightarrow \text{for all } y \in \Sigma^n$$

Case 2: $S \neq 0^n$

There are two strings $x \in f^{-1}(\{z\})$ for each $z \in \text{range}(f)$, if $w \in f^{-1}(\{z\})$ is either one of them then $w \oplus S$ is the other

$$\sum_{x \in f^{-1}(\{z\})} \left| (-1)^{x \cdot y} \right|^2 = \left| (-1)^{w \cdot y} + (-1)^{(w \oplus S) \cdot y} \right|^2$$

